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3rd Sem / Mechanical Engg, Mechanical(Tool & Die Design)Agri, Prod, CNC, CAD/CAM, GE, Metallurgy, Pack. Tech, Print Making Tech., Mech (Ad. Manu. Tech.), Mech Engg (Fabrication Tech)

Subject : Mechanical Engineering Drawings - II / Mechanical Engineering Drawing / Machine Drawing

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 What is the purpose of LIMITS in production parts?
- a) To ensure a perfect fit in all the size cases
 - b) To allow for variation in assembled part dimensions
 - c) To specify the exact size of a part
 - d) To set an upper size boundation of the part.
- Q.2 Which type of motion is easily transmitted through universal coupling?
- a) Axial motion
 - b) Rotational motion
 - c) Axial and rotational motion
 - d) It cannot transmit any type of motion

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- Q.3 A tolerance of ± 0.05 mm means that
- a) The dimension can vary upto 0.05 mm in a job
 - b) The dimension can vary upto 0.10 mm in a job
 - c) The dimension can vary between + 0.05 mm to -0.05 mm in a job
 - d) Both b and c
- Q.4 Which of the following is not a part of nomenclature of gear?
- a) Involute profile
 - b) Addendum
 - c) Pitch circle
 - d) Dedendum
- Q.5 In a shaft-hole system, which fit has shaft size greater than the hole size?
- a) Force fit
 - b) Clearance fit
 - c) Interference fit
 - d) Running fit
- Q.6 A circle will appear as _____ on an isometric drawing.
- a) Ellipse
 - b) Cycloid
 - c) Circle
 - d) Parabola

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 In _____ fit, the smallest shaft is always larger than the largest hole.
- Q.8 _____ coupling has shafts at parallel axis offset.
- Q.9 _____ gears have teeth at angles to the axis of rotation.
- Q.10 Function of flywheel is to _____

Q.11 Material of balls in ball bearings is _____.

Q.12 Define Bilateral Tolerance.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain various types of fits with diagram.

Q.14 Enlist the various parts of screw jack and give their function & respective materials.

Q.15 Draw a free hand front view of Universal coupling.

Q.16 Explain the function of bearing and how roller bearing achieves its purpose?

Q.17 What is the purpose of bushes in bushed bearing, give their material and the function of oil bushes.

Q.18 Draw a free hand front view of a Piston giving its material.

Q.19 What is a crank shaft? Give its material and the method of manufacturing it.

Q.20 What is the use of steam stop valve & Blow off cock.

Q.21 Draw free hand sketch of one view of a typical wall bracket.

Q.22 Explain the working of expansion joint, its parts and their materials with a diagram.

SECTION-D

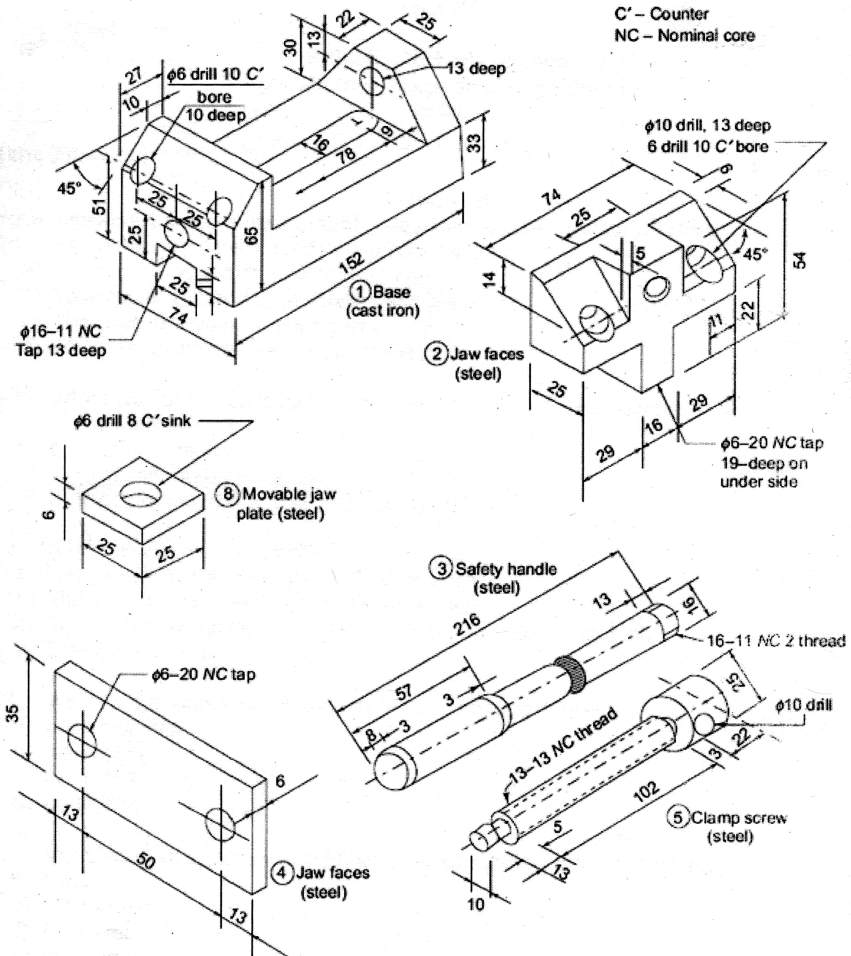
Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Draw two free hand views of Flanges pipe joint assembly.

Q.24 Classify various pipe joints and draw their symbols.

Q.25 Assemble the drill machine vice as shown in figure below and draw

- a) elevation
- b) Plan view
- c) Bill of material



- Dimension units may be taken in mm.
- Assume suitable dimensions are required if not specified or not visible clearly in the figure.

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